

Operational Decisions Timeline (18-Month Horizon)

Month 0 — Opening Conditions

- Started at 95% baseline capacity, supported by 30 days of beginning inventory, giving us a buffer to absorb early demand.

Months 1–4 — Stable Operations

- Held capacity flat at 95%, intentionally relying on opening inventory rather than over-producing ahead of uncertainty.

Months 5–7 — Supply Shock Period

- Maintained 95% capacity during the shock, choosing stability over reactive ramp-ups while assessing the magnitude of disruption.
- Allowed backlog to accumulate, using the backlog signal to quantify the true supply-demand imbalance.

Month 8 — Post-Shock Stabilization

- Restored capacity to 100% once the shock ended, stabilizing supply while monitoring backlog decay and aging behavior.

Month 9 — Controlled Recovery Ramp

- Executed a planned ramp to 110% capacity, accelerating recovery without risking an inventory overshoot.

Months 10–11 — Backlog Quality Monitoring

- Tracked aging buckets and risk distribution, using backlog quality to determine whether additional intervention was needed.

Month 12 — Targeted Year-End Surge

- Deployed a short, targeted surge (120% capacity) to eliminate remaining backlog before year-end.

Months 13–18 — Return to Steady State

- Tapered capacity back toward 100%, avoiding overproduction and returning the system to a stable operating rhythm.
- Closed the period with a controlled inventory position, having cleared backlog without creating excess stock.